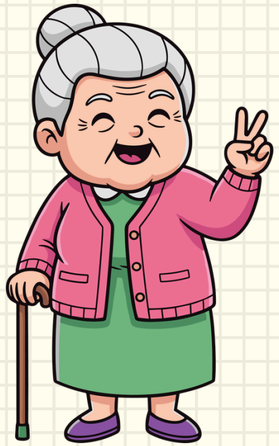
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How Can Telehealth Services Increase Repeat Visits Among Older Adults?

PROBLEM STATEMENT

BACKGROUND

- Although the world is technologically advancing and better technologies equipped for healthcare services exist, older adults are generally **apprehensive** when it comes to using these technologies.
- Its growing presence reflects changing healthcare needs, especially in societies with a larger proportion of older adults who may require more frequent medical attention and long-term care support.

Rationale

- This topic was chosen because it reflects a current healthcare challenge with clear implications for both patients and providers.
- Through this analysis, our team strives to identify the factors that affect usage and henceforth provide relevant solutions.



CURRENT CONDITION OF TELEHEALTH SERVICES IN SINGAPORE

- **Modern pre-COVID Singapore**: Tele-treatment had the least traction due to face-to-face consultations being the preferred norm, with clinicians often emphasising the importance of physical examinations (Low & Low, 2025).
- **Post COVID Singapore**: The industry quickly pivoted to the provision of telemedicine for acute care, chronic disease management, mental health support and home medicine delivery services to meet the demand (Low & Low, 2025).
- **Healthier SG**: Singapore has also embarked on a nationwide effort called Healthier SG to shift care from hospitals to homes and communities. Technology is a key aspect that's making this possible, through apps like HealthHub and Healthy 356 (NUS, 2025).
- **Generative AI**: Singapore's Ministry of Health is also looking into scaling the use of generative AI tools to automate tasks, like documentation and summarisation of medical records. (NUS ISS, 2025).

BENEFITS OF TELEHEALTH SERVICES

- **Overcrowding**: Tackle the problem of Singapore's overcrowded hospitals and specialist outpatient clinics, while patients benefit as they save time on travelling and waiting for appointments. (Synapxe, 2021).
- **Growing Potential**: Potential to improve patient outcomes and access to care is multifaceted by solving various challenges, (Ezeamii et al., 2024).
- **Coordination**: Enhances care coordination by allowing easier follow-ups and communication between patients and providers.



POSSIBLE CAUSES

- Low Digital Literacy
- Preference for Face-Face Consultations
- Limited Trust in Telehealth Medicine



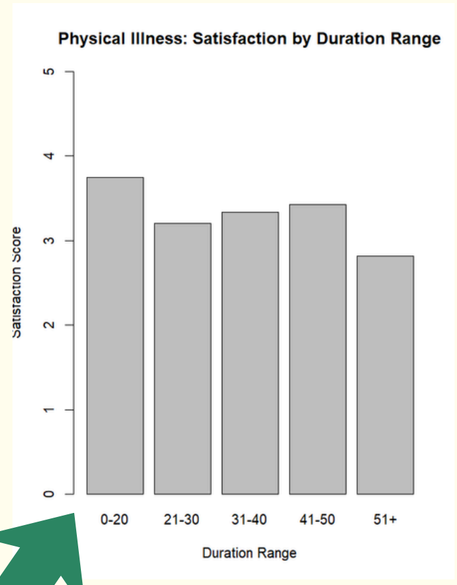
RECOMMENDATIONS

- Simplify telemedicine platform design
- Ensure follow up visits are prioritised.
- Build trust in remote consultations



Keeping these causes and recommendations in mind, we now use statistical analysis to further evaluate these as well as various other factors.

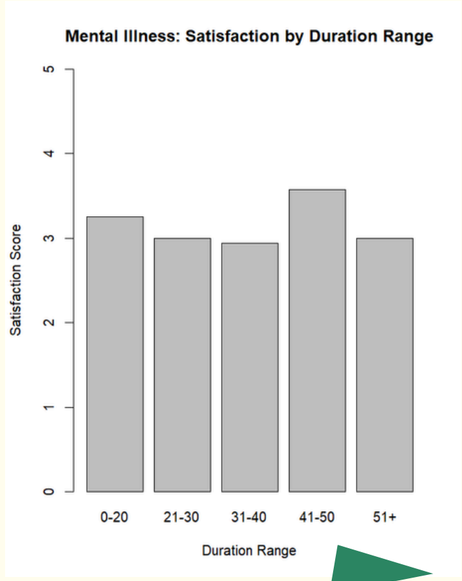
MEAN



Physical illness

Asthma
Diabetes
Hypertension

Highest mean satisfaction score lies in the duration range **0-20 minutes**



Mental illness

Anxiety
Depression

Highest mean satisfaction score lies in the duration range **41-50 minutes**

- **Optimal** consultation duration for physical illness is between 0-20 minutes.
- **Optimal** consultation duration for mental illness is between 41-50 minutes

MEAN

Implications

- **Duration of the consultation matters : to properly diagnose & tend to the needs of patients**
- **Different optimal consultation duration depending on the nature of the illness**

Solutions

Impose consultation duration guidelines for the doctors

- Ensure efficiency while maintaining the accuracy & quality of service
- Extend the consultation session only when necessary

HYPOTHESIS TESTING

Service Type

Satisfaction Comparison: General vs Follow-up Consultations

Let **X** be the **satisfaction score** of **General consultations**
Let **Y** be the **satisfaction score** of **Follow-up visits**

$H_0: \mu_1 - \mu_2 = 0$

$H_1: \mu_1 - \mu_2 \neq 0$

```
        Welch Two Sample t-test

data:  X and Y
t = -2.49, df = 97.391, p-value = 0.01447
alternative hypothesis: true difference in means is not equal to 0
90 percent confidence interval:
 -0.6370545 -0.1272972
sample estimates:
mean of x mean of y
 2.957447  3.339623
```

Analysis

At a **90% confidence level**, the **p-value < 0.10**, so there is strong evidence to **reject H_0**. This suggests a **statistically significant difference** in satisfaction levels between General consultations and Follow-up visits. Furthermore, **Follow-up visits have higher mean satisfaction scores than General consultations**, indicating areas for **improvement** in the **latter**.

HYPOTHESIS TESTING

Service Type

Satisfaction Comparison: General vs Follow-up Consultations

Solutions

Healthcare providers should **prioritise optimising General consultations** to **close the satisfaction gap**. This can be achieved by:

- **Increasing specialist manpower** to reduce patient waiting times
- **Improving workflow efficiency** for smoother consultations
- **Providing additional patient support**, such as guidance or digital tools

Implications

- **Increased patient trust and satisfaction**
- **Better health outcomes**
- **Stronger service quality and reputation**

Closing the satisfaction gap helps providers deliver a **more consistent, patient-centered experience** across **all service types**.

HYPOTHESIS TESTING Cost

To determine if TeleConsult cost affects satisfactions of telehealth consultation. Set confidence level as 90%.

Healthcare cost

Let X be the satisfaction score of individuals whose TeleConsult cost <= $150 and below. Let Y be the satisfaction score of individuals whose TeleConsult cost > $150.

H0: $\mu_X - \mu_Y = 0$
H1: $\mu_X - \mu_Y \neq 0$

```
        Welch Two Sample t-test

data:  X and Y
t = -1.1118, df = 53.046, p-value = 0.2712
alternative hypothesis: true difference in means is not equal to 0
90 percent confidence interval:
 -0.8798785  0.1775958
sample estimates:
mean of x mean of y
 2.620690  2.971831
```

> sig. level 0.1

Insurance

Let X be the satisfaction score of individuals whose insurance status is “Insured”. Let Y be the satisfaction score of individuals whose insurance status is “Not insured”.

H0: $\mu_X - \mu_Y = 0$
H1: $\mu_X - \mu_Y \neq 0$

```
        Welch Two Sample t-test

data:  X and Y
t = 0.59179, df = 97.034, p-value = 0.5554
alternative hypothesis: true difference in means is not equal to 0
90 percent confidence interval:
 -0.1678890  0.3537864
sample estimates:
mean of x mean of y
 3.208333  3.115385
```

> sig. level 0.1

In both tests, p-value greater than significance level 10%.

- Marginal evidence to reject two null hypotheses.

HYPOTHESIS TESTING — Cost

To determine if TeleConsult cost affects satisfactions of telehealth consultation. Set confidence level as 90%.

Healthcare cost

Let X be the satisfaction score of individuals whose TeleConsult cost <= $150

```
data:
t = -1.1
alternati
90 percent
 -0.8798785  0.1
sample estimates:
mean of x mean
 2.620690  2.971831
```

Insurance

Let X be the satisfaction score of individuals whose satisfaction score

```
3.208333  3.115385
```

Healthcare cost and insurance status are statistically insignificant to patients' satisfaction to their TeleConsult.

p-value greater than significance level 10%.

- Marginal evidence to reject null hypothesis.
- The two factors tested, healthcare cost and insurance, are **statistically insignificant** to patient's **satisfaction** to the TeleConsult.

HYPOTHESIS TESTING

Cost

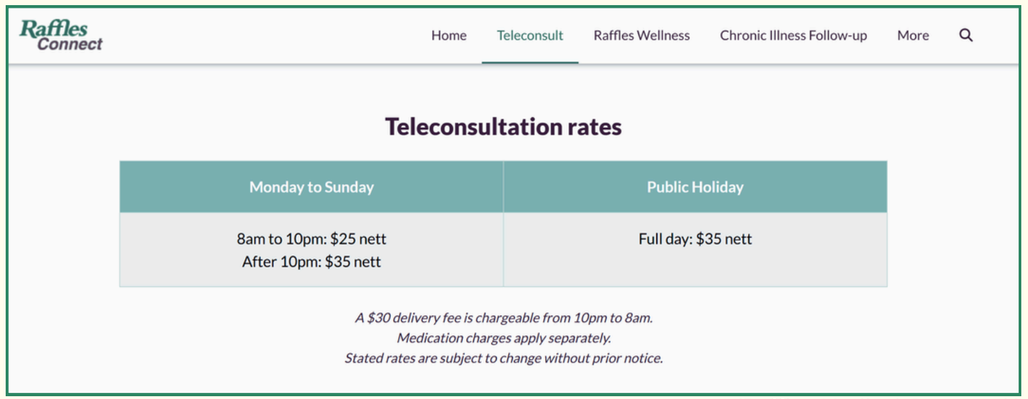
Analysis

These **two factors**, healthcare cost and insurance status, **does not seem to be affect satisfaction** score.

- Amount of out-of-pocket payment is not a significant factor towards patients' satisfaction score.

Possible reason:

- Patients have knowledge of the price and are already **prepared to pay a certain amount**. E.g. They can find out the expected rates on the TeleConsult website.
  - What affects their satisfaction is the quality of the consultation itself.
  - If quality of the consult was not that good, perhaps pricing becomes a more significant factor to satisfaction as patient cannot justify the high price point of their consult.



(Raffles Medical Group, nd.)

HYPOTHESIS TESTING

Technical Issues

Satisfaction Comparison: General vs Follow-up Consultations

$H_0: \mu_1 - \mu_2 = 0$

$H_1: \mu_1 - \mu_2 \neq 0$

- μ_1: Sample mean of satisfaction scores for # of technical Issues ≤ 5
- μ_2: Sample mean of satisfaction scores for # of technical Issues ≥ 6

```
        Welch Two Sample t-test

data:  x and y
t = 3.6025, df = 43.183, p-value = 0.0008093
alternative hypothesis: true difference in means is not equal to 0
90 percent confidence interval:
 0.3282493 0.9025200
sample estimates:
mean of x mean of y
 3.615385  3.000000
```

Analysis

At 90% confidence level, p-value < 0.01, we have strong evidence to reject H_0. Number of technical issues is statistically significant to satisfaction levels, with a higher number of technical issues negatively affecting user satisfaction.

HYPOTHESIS TESTING

Technical Issues

Result comparison

Implications



- Technical issues are one of the key problems of user experience
- Poor user experience may reduce trust and continued use of telehealth platforms

Solutions

To improve the current technical issues,

1. Simplify interface design
2. Provide technical support (e.g. AI live chat)
3. Enhance digital literacy



REGRESSION ANALYSIS

```
Residuals:
     Min       1Q   Median       3Q      Max
-1.58299 -0.40803  0.07393  0.45332  1.41388

Coefficients:
                           Estimate Std. Error t value Pr(>|t|)
(Intercept)                5.846727   1.205270   4.851 5.19e-06 ***
TechnicalIssues           -0.140733   0.036185  -3.889 0.000194 ***
Duration                  -0.009118   0.005062  -1.801 0.075019 .
HealthcareCost            -0.003092   0.001341  -2.306 0.023455 *
GenderMale                -0.282788   0.142680  -1.982 0.050568 .
ServiceTypeGeneral Consult -0.209232   0.148677  -1.407 0.162825
TelehealthPlatformTeams    0.174632   0.174929   0.998 0.320839
TelehealthPlatformZoom     0.581913   0.167443   3.475 0.000791 ***
InsuranceTypeNot insured  -0.113549   0.138244  -0.821 0.413633
TypesofIllnessPhysical     0.086776   0.143907   0.603 0.548041
Age                       -0.013023   0.015537  -0.838 0.404161
```

- From the result, we can see that only number of technical issues and telehealth platform (zoom) show statistically significant results at 10% sig. level
- With their beta coefficient being -0.1407 and 0.5819 respectively
- With everything else being equal:
  - 1 unit increase in technical issues experienced will decrease the satisfaction score by 0.1407
  - 1 unit change in zoom platform usage increase the satisfaction score by 0.5819

Multiple Regression model

SatisfactionScore = b0 + b1(TechnicalIssues) + b2(Duration) + b3(HealthcareCost) + b4(GenderMale) + b5(ServiceTypeGeneralConsult) + b6(TelehealthPlatformTeams) + b7(TelehealthPlatformZoom) + b8(InsuranceTypeNot insured) + b9(TypesofIllnessPhysical) + b10(Age) + e

REGRESSION ANALYSIS

Result comparison

Implication

1. **Technical issues are the main barrier to repeat visits**
2. **Platform simplicity strongly influences user retention**
3. **User experience matters more than cost or demographics**

Solution after regression analysis

Leverage Zoom like platform strengths:

1. **Standardize on “Zoom-like” experience**
2. **Use widely familiar platforms by integrating zoom and avoid complex systems**
3. **Reduce onboarding friction**

CONCLUSION

To improve repeat visit rates among elderly users, telehealth providers should prioritise reducing technical barriers and adopting simple, familiar platforms such as Zoom. They are also recommended to prioritise optimising General consultations to close the satisfaction gap.

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